

MATH 1342 – Elementary Statistics – Week 12: Inference for a Mean

Nov 9 – 11 | 6.2 Inference for a Mean (Distribution, CI, Hypothesis Test) • Veterans Day – No Class (Wed Nov 11) | Fall 2026 • Section 019 • MW 7:30–8:50 PM • RRC Room 8210

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
4. Find and use theoretical probabilities from normal, t, chi-squared and F distributions to form confidence intervals and test hypotheses. Apply the 95% rule to normal and to approximately normal distributions. 8. Use the sampling distributions of sample proportions and sample means to answer appropriate questions. 9. Estimate single means, difference of two means, single proportions and difference of two proportions using confidence intervals. Interpret the results. 10. Demonstrate skills in hypothesis testing for means and proportions, for single populations and comparison of two populations.	1. Explain why inference for a mean uses the t distribution with $df = n - 1$ instead of the normal distribution. (CO4) 2. Check the conditions for t procedures ($n \geq 30$, or roughly normal data without extreme outliers). (CO4, CO8) 3. Construct and interpret a t confidence interval for a population mean, including the margin of error. (CO4, CO9) 4. Carry out a one-sample t test for μ and state the conclusion in context. (CO4, CO10)	e-Text: 6.2-CI Confidence Interval for a Mean	All	Participation: Inference for a Mean – gribbles CI and Chips Ahoy test worked in groups	MO2, MO3, MO4	Online Homework: 6.2 Inference for a Mean (due Sun Nov 15, 11:59 PM)	All
		e-Text: 6.2-D Distribution of a Mean	All				
		e-Text: 6.2-HT Hypothesis Test for a Mean	All				
		Interactive Lecture: 6.2 Inference for a Mean	All				
		Slides: 6.2 Inference for a Mean	All				
		StatKey – Theoretical Distributions (t)	MO3, MO4				